Thadomal Shahani Engineering College, Bandra
Department of Artificial Intelligence & Data Science

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TE - 1	**Batch: T11**	**Sem: 5**	**Roll No:12**	**Date:4/9/2026**

Experiment 06

AIM: To study and Implement Agile Software Project lifecycle using Jira with DevOps Pipeline Integration.

LO Mapped : 01

LO : Apply Agile methodologies and use Jira for sprint planning, task tracking, and integrating with DevOps workflows.

THEORY:

What is jira?

Jira is an industry-standard project management and issue-tracking software developed by Atlassian. It is widely used by software engineering, DevOps, and business teams to plan, track, and manage work throughout the software development life cycle.

Core Features of Jira

- Agile Framework Support: Built-in support for Scrum and Kanban methodologies, allowing teams to create backlogs, plan sprints, and organize work items on visual boards.
- Issue & Task Tracking: Tracks individual units of work—such as user stories, tasks, bugs, and epics—using unique issue keys (e.g., SCRUM-1).
- DevOps & Tool Integrations: Connects directly with version control systems (like GitHub or Bitbucket) to link branches, commits, and pull requests directly to project tasks.
- Real-time Dashboards & Reporting: Generates continuous visibility, velocity charts, and release timelines to track team productivity and project milestones**.**

1. Overview & Theoretical Background

Modern DevOps & Agile Methodologies

DevOps bridges project management with technical execution. By pairing Agile planning tools with Version Control Systems (VCS), software teams maintain full visibility across the Software Development Life Cycle (SDLC).

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Atlassian Jira Platform

Jira is an enterprise project management and issue-tracking application developed by Atlassian. Built to support Agile frameworks (Scrum and Kanban), it organizes work into trackable units called issues or user stories. Every work item is assigned a unique key (such as SCRUM-1), which serves as a global tracking ID across the development pipeline.

Version Control & Feature Branch Workflows

Udit

The global Development hub in Jira was reviewed to verify that repository Yogini21-jpg/ecommerce-app was synced and PR #2 was logged as MERGED.

> Figure 6: Global Jira Development dashboard verifying end-to-end sync for SCRUM-1.sing Git alongside remote platforms like GitHub allows developers to isolate code changes. Under a feature-branching strategy, developers create dedicated branches named after specific task identifiers (e.g., SCRUM-1-edit-catalog-title). Modifications are tested in isolation and submitted via Pull Requests (PRs) before being reviewed and merged into the primary production branch (main).

Automated Cross-Platform Traceability

Integrating Jira Cloud with GitHub via OAuth webhooks enables automated metadata synchronization:

- Smart Parsing: Referencing a Jira issue key inside commit messages, branch names, or PR titles automatically maps those Git actions to the corresponding Jira ticket.
- Bi-Directional Updates: Code activities—including branch creation, commit history, and PR merge statuses—are updated live inside Jira's issue panels and central development dashboards without manual intervention.

2. Practical Execution & Image Mapping

Step 1: Organization & Team Environment

Collaborators were added to the central Atlassian workspace to enable role-based access control and task delegation across team members (Yogini, Gyan Achantani, Rashi Bhatia, and Saniya Bijlani).

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Figure 1: Atlassian People directory confirming verified team members in the project workspace.

Step 2: Sprint Scheduling & Planning

Tasks were scheduled and assigned across the sprint timeline to set feature ownership and target delivery dates.

Figure 2: Jira Timeline view showing assigned user stories, workflow statuses, and sprint schedules.

Step 3: Product Backlog Configuration

Core requirements were structured as user stories within the backlog. Task SCRUM-1 (Sheify Website Development) was staged for development.

Figure 3: Product backlog containing user story SCRUM-1 staged for sprint execution.

Step 4: Branching, Commits & PR Merging on GitHub

A dedicated feature branch SCRUM-1-edit-catalog-title was created on GitHub. Changes to index.html were committed using key SCRUM-1. Pull Request #2 was opened by bhatiarashi19-ops and merged into main.

Figure 4: GitHub PR #2 associated with issue key SCRUM-1 successfully merged into main.

Step 5: Jira Issue Linkage & Panel Audit

Including SCRUM-1 in the Git workflow automatically populated the Development panel inside Jira ticket SCRUM-1, recording 1 branch, 2 commits, and 1 merged PR.

Figure 5: Jira work item SCRUM-1 displaying real-time development metrics synced from GitHub.

Step 6: Global Integration Audit

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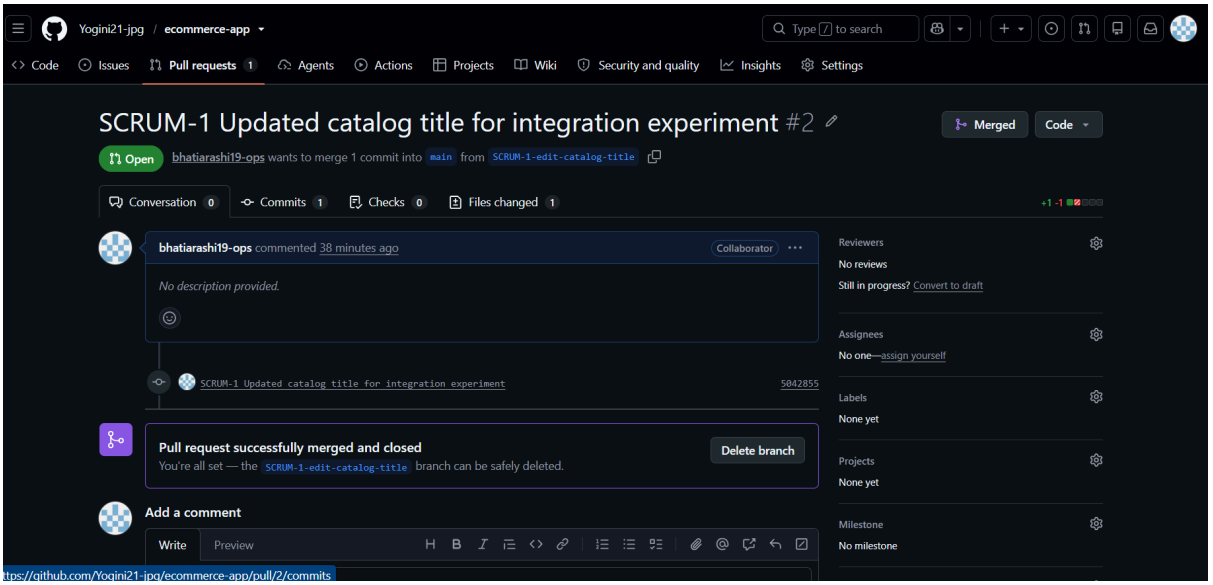
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The global Development hub in Jira was reviewed to verify that repository `Yogini21-jpg/ecommerce-app` was synced and PR #2 was logged as **MERGED**.

Figure 6: Global Jira Development dashboard verifying end-to-end sync for SCRUM-1.

```
PS C:\Users\Bhati\ecommerce-app> git push origin SCRUM-1-edit-catalog-title
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 389 bytes | 389.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'SCRUM-1-edit-catalog-title' on GitHub by visiting:
remote:      https://github.com/Yogini21-jpg/ecommerce-app/pull/new/SCRUM-1-edit-catalog-title
remote:
To https://github.com/Yogini21-jpg/ecommerce-app.git
 * [new branch]      SCRUM-1-edit-catalog-title -> SCRUM-1-edit-catalog-title
PS C:\Users\Bhati\ecommerce-app> _
```

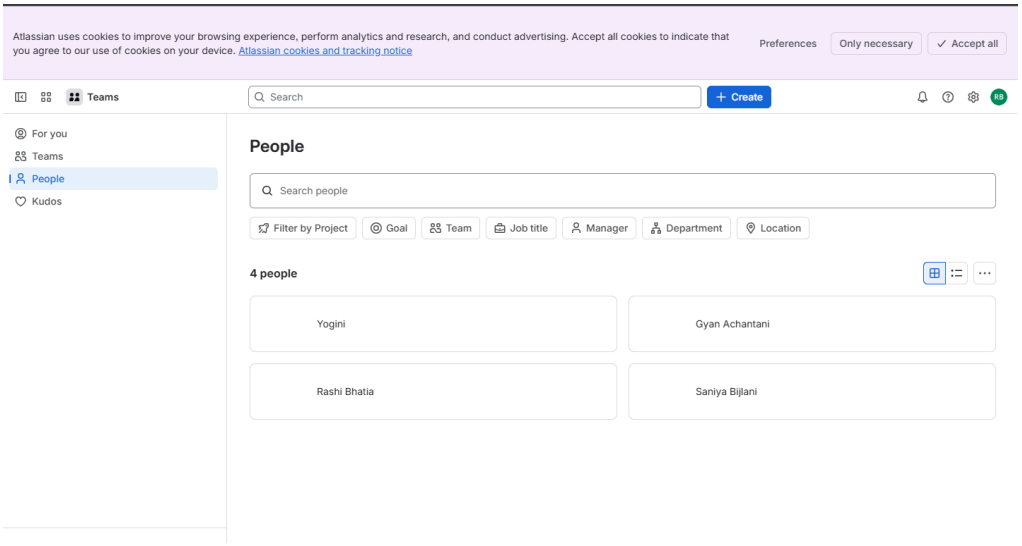
Section: Jira Issue Linkage & Development Panel Verification



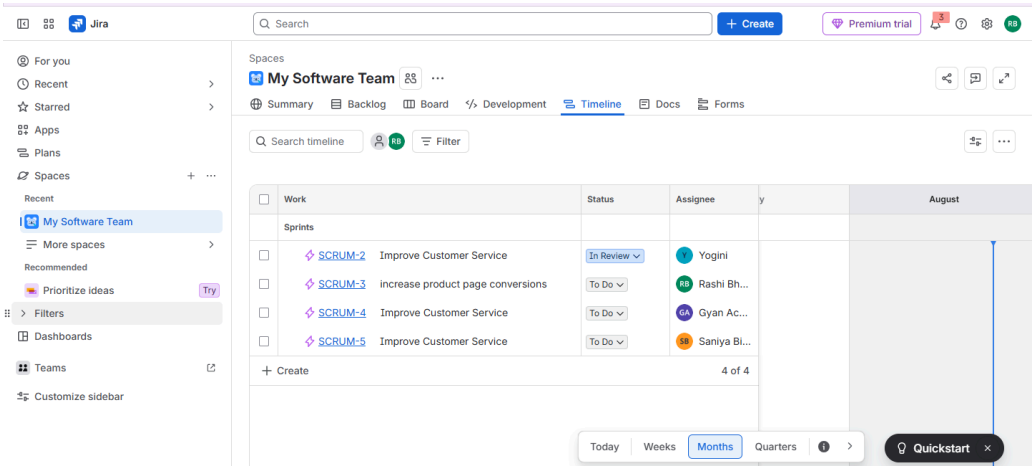
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Section: Organization Setup & Team Collaboration Environment



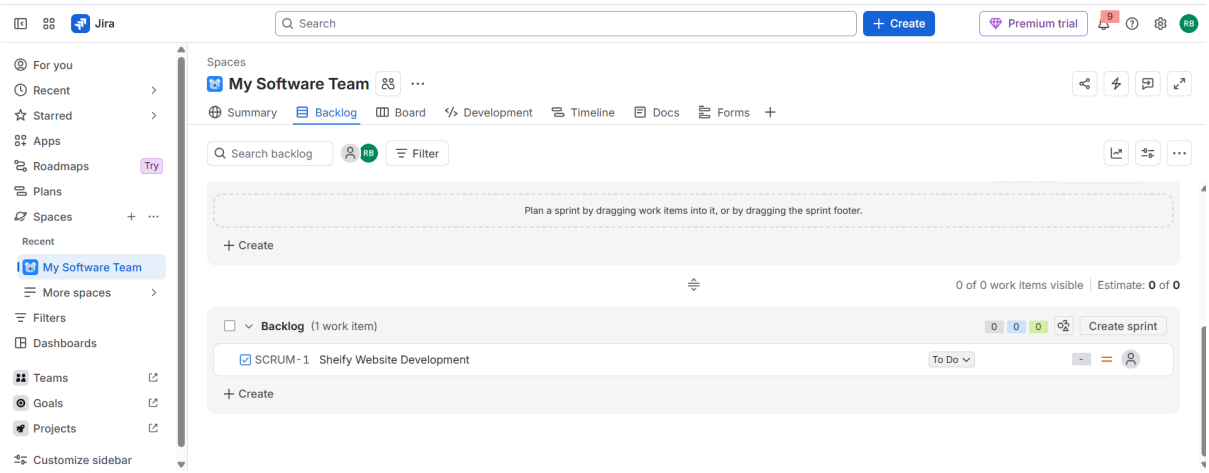
Section: Sprint Timeline & Task Allocation



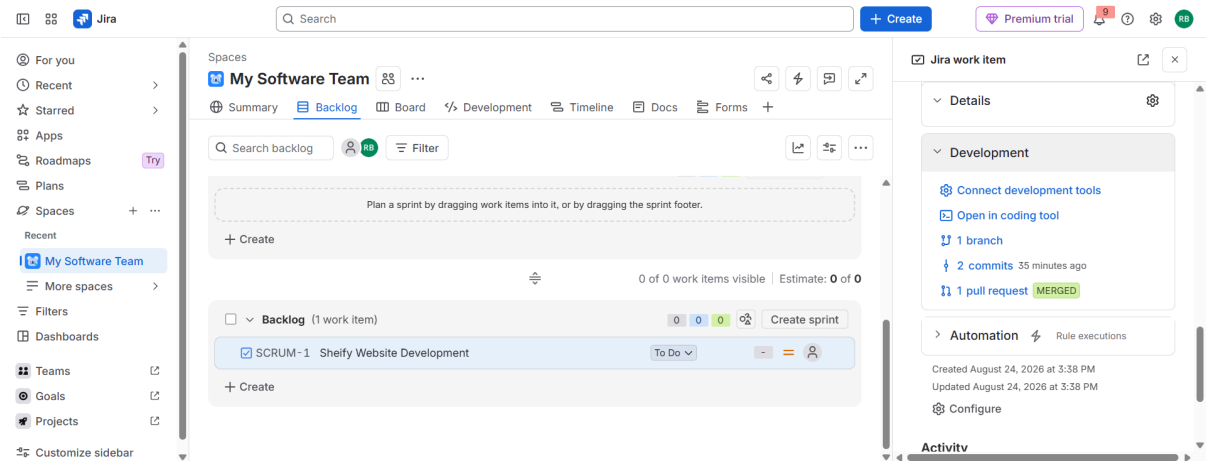
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Section: Global Integration Verification & Audit Trail



Section: Jira Issue Linkage & Development Panel Verification



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Conclusion

The integration between GitHub and Jira Cloud was successfully configured, executed, and verified:

- Workspace Setup: User stories and sprint schedules were defined and assigned across team members in Jira.
- Isolated Execution: Code updates were safely developed in dedicated feature branches and merged into `main` via GitHub PR `#2`.
- Automated Sync: Linking issue key `SCRUM-1` across Git commits and branches automatically updated Jira's tracking views and global dashboards.

This automated pipeline removes manual status reporting, provides a clear audit history, and maintains full visibility across the development process.